

A Short Guide to Lemma D.5

The tangent witness for the multimode Gaussian fidelity bound

Abstract

Lemma D.5 converts the product stochastic inequality of Lemma D.4 into the sharp multimode root-fidelity bound needed for Gaussian amplification. Its key step is a tangent-line inequality for the square root. The tangent is taken at the photon-number law of the quantum-limited amplifier, and its slope is exactly a bounded product of decreasing weights, so Lemma D.4 controls the linear correction. This note derives that step explicitly and separates it from the external least-noise theorem used earlier in the dependency chain.

1 Purpose of the lemma

At the origin, the desired amplified state is the original product thermal state Φ_0 , while every physical amplifier of gain $\gamma > 1$ must add noise. The quantum-limited amplifier produces a hotter product thermal state $\tau^{(\gamma)}$. Lemma D.5 proves that no displacement- and phase-covariant operation, even a trace-decreasing one, can have larger root fidelity with Φ_0 than this quantum-limited output.

The proof has a useful convex-analytic form. Root fidelity with a fixed diagonal target is a concave function of the output probabilities. Its tangent plane at the quantum-limited distribution is a linear photon-number test. Earlier stochastic-order lemmas show that every allowed output lies on the correct side of that test.

2 Background

2.1 Root fidelity for diagonal positive operators

For positive trace-class operators R, S , possibly subnormalized, the paper uses root fidelity

$$B(R, S) = \text{Tr} \sqrt{R^{1/2} S R^{1/2}}.$$

If R and S are diagonal in the same basis, with masses $P = (P_{\mathbf{k}})$ and $t = (t_{\mathbf{k}})$, then

$$B(R, S) = B(P, t) := \sum_{\mathbf{k}} \sqrt{P_{\mathbf{k}} t_{\mathbf{k}}}. \quad (1)$$

This is the classical Hellinger affinity. The formula remains valid when $\sum_{\mathbf{k}} P_{\mathbf{k}} < 1$.

For tensor products, root fidelity is multiplicative:

$$B\left(\bigotimes_{\ell} R_{\ell}, \bigotimes_{\ell} S_{\ell}\right) = \prod_{\ell} B(R_{\ell}, S_{\ell}). \quad (2)$$

2.2 Thermal laws and the one-mode factor

For $0 < q < 1$,

$$\tau_q = (1 - q) \sum_{k \geq 0} q^k |k\rangle\langle k|.$$

At intensity gain $\gamma > 1$, define

$$q^{(\gamma)} = 1 - \frac{1 - q}{\gamma}. \quad (3)$$

The quantum-limited amplifier sends τ_q to $\tau_{q^{(\gamma)}}$. Summing a geometric series gives

$$\begin{aligned} b_\gamma(q) &:= B(\tau_{q^{(\gamma)}}, \tau_q) \\ &= \frac{\sqrt{(1 - q)(1 - q^{(\gamma)})}}{1 - \sqrt{qq^{(\gamma)}}} = \frac{\sqrt{\gamma} + \sqrt{q(\gamma - 1 + q)}}{\gamma + q}. \end{aligned} \quad (4)$$

The last expression follows by substituting (3) and rationalizing the denominator.

For r modes, set

$$\Phi_0 = \bigotimes_{\ell=1}^r \tau_{q_\ell}, \quad \tau^{(\gamma)} = \bigotimes_{\ell=1}^r \tau_{q_\ell^{(\gamma)}}. \quad (5)$$

Their joint number distributions are

$$t_{\mathbf{k}} = \prod_{\ell=1}^r (1 - q_\ell) q_\ell^{k_\ell}, \quad (6)$$

$$t_{\mathbf{k}}^{(\gamma)} = \prod_{\ell=1}^r (1 - q_\ell^{(\gamma)}) (q_\ell^{(\gamma)})^{k_\ell}. \quad (7)$$

2.3 Why the output is diagonal

Suppose Γ is covariant under independent phase rotations in every mode. Since Φ_0 is invariant under those rotations, so is $\Gamma(\Phi_0)$. The joint eigenspaces of the independent number operators $(\hat{N}_1, \dots, \hat{N}_r)$ are one-dimensional. Therefore

$$\Gamma(\Phi_0) = \sum_{\mathbf{k}} P_{\mathbf{k}} |\mathbf{k}\rangle\langle \mathbf{k}|, \quad c := \sum_{\mathbf{k}} P_{\mathbf{k}} \leq 1. \quad (8)$$

Independent phase covariance is stronger than covariance under only the total phase rotation; the latter has degenerate eigenspaces and would not by itself force diagonality in the joint number basis.

3 The input from Lemma D.4

Lemma D.4 says that for every bounded product weight

$$W(\hat{\mathbf{N}}) = \prod_{\ell=1}^r w_\ell(\hat{N}_\ell),$$

with each w_ℓ nonnegative and decreasing,

$$\sum_{\mathbf{k}} P_{\mathbf{k}} w_{\mathbf{k}} \leq c \sum_{\mathbf{k}} t_{\mathbf{k}}^{(\gamma)} w_{\mathbf{k}}, \quad w_{\mathbf{k}} = \prod_{\ell} w_\ell(k_\ell). \quad (9)$$

This is a conic inequality because it includes the output mass c .

Lemma D.4 ultimately depends on the Guță–Matsumoto least-noise stochastic-order theorem through Lemma D.3. That external theorem is not invoked anew in D.5. The choice of weight, tangent calculation, and fidelity argument below are all proved within the manuscript.

4 Statement of Lemma D.5

Lemma D.5 (rephrased). *Under the assumptions and notation of Lemma D.4,*

$$B(\Gamma(\Phi_0), \Phi_0) \leq \prod_{\ell=1}^r b_\gamma(q_\ell). \quad (10)$$

Equality is attained by the product quantum-limited amplifier.

5 The tangent inequality

The scalar function $x \mapsto \sqrt{x}$ is concave, so its tangent line at $y > 0$ lies above its graph. In the scaled form needed here, for $x \geq 0$ and $s, y > 0$,

$$\sqrt{xs} \leq \sqrt{ys} + \frac{1}{2} \sqrt{\frac{s}{y}} (x - y). \quad (11)$$

There is also a direct identity for the gap:

$$\sqrt{ys} + \frac{1}{2} \sqrt{\frac{s}{y}} (x - y) - \sqrt{xs} = \frac{1}{2} \sqrt{\frac{s}{y}} (\sqrt{x} - \sqrt{y})^2 \geq 0. \quad (12)$$

Apply (11) separately at every $\mathbf{k}$ with

$$x = P_{\mathbf{k}}, \quad y = t_{\mathbf{k}}^{(\gamma)}, \quad s = t_{\mathbf{k}}.$$

All thermal masses are strictly positive. Define the tangent slope, apart from the common factor $1/2$, by

$$w_{\mathbf{k}} := \sqrt{\frac{t_{\mathbf{k}}}{t_{\mathbf{k}}^{(\gamma)}}}. \quad (13)$$

Summing the pointwise tangent inequalities gives

$$B(P, t) \leq B(t^{(\gamma)}, t) + \frac{1}{2} \sum_{\mathbf{k}} w_{\mathbf{k}} (P_{\mathbf{k}} - t_{\mathbf{k}}^{(\gamma)}). \quad (14)$$

This is the “tangent inequality for the square root” in the paper. There is no hidden operator inequality: it is the elementary scalar inequality (11), applied after phase covariance has made all states diagonal.

The infinite sums are legitimate. The weights will be bounded, so the two weighted sums in the correction term are absolutely convergent; the fidelity sums are finite by Cauchy–Schwarz.

6 Why the tangent slope is an admissible stochastic weight

Using (6)–(7), the weight factorizes:

$$w_{\mathbf{k}} = \prod_{\ell=1}^r w_{\ell}(k_{\ell}), \quad (15)$$

$$w_{\ell}(k) = \sqrt{\frac{1 - q_{\ell}}{1 - q_{\ell}^{(\gamma)}}} \left(\frac{q_{\ell}}{q_{\ell}^{(\gamma)}} \right)^{k/2}. \quad (16)$$

Moreover,

$$q_{\ell}^{(\gamma)} - q_{\ell} = \frac{(\gamma - 1)(1 - q_{\ell})}{\gamma} > 0.$$

Hence $0 < q_{\ell}/q_{\ell}^{(\gamma)} < 1$, and every $w_{\ell}(k)$ is bounded, positive, and decreasing. Thus (15) belongs exactly to the product class controlled by Lemma D.4.

This matching is the central design of the proof: the derivative of fidelity at the quantum-limited output happens to be a decreasing product observable.

7 Proof walkthrough

Step 1: apply Lemma D.4 to the tangent slope

For the weight (13), equation (9) gives

$$\sum_{\mathbf{k}} w_{\mathbf{k}} P_{\mathbf{k}} \leq c \sum_{\mathbf{k}} w_{\mathbf{k}} t_{\mathbf{k}}^{(\gamma)}. \quad (17)$$

Because $c \leq 1$ and every summand is nonnegative,

$$\sum_{\mathbf{k}} w_{\mathbf{k}} (P_{\mathbf{k}} - t_{\mathbf{k}}^{(\gamma)}) \leq (c - 1) \sum_{\mathbf{k}} w_{\mathbf{k}} t_{\mathbf{k}}^{(\gamma)} \leq 0. \quad (18)$$

This also explains why allowing trace-decreasing maps causes no problem: lost mass only makes the tangent correction smaller.

Step 2: return to the fidelity

Insert (18) into (14):

$$B(P, t) \leq B(t^{(\gamma)}, t). \quad (19)$$

By (1), the left side is $B(\Gamma(\Phi_0), \Phi_0)$. By multiplicativity and (4),

$$B(t^{(\gamma)}, t) = \prod_{\ell=1}^r B(\tau_{q_{\ell}^{(\gamma)}}, \tau_{q_{\ell}}) = \prod_{\ell=1}^r b_{\gamma}(q_{\ell}). \quad (20)$$

Equations (19) and (20) prove (10).

Step 3: check sharpness

The product quantum-limited amplifier is trace preserving and has the required displacement and phase covariance. It maps

$$\Phi_0 \longmapsto \tau^{(\gamma)},$$

so its fidelity is exactly the right side of (20). Therefore the bound is attained.

8 A one-mode numerical illustration

Take $q = 1/2$ and $\gamma = 2$. Then $q^{(\gamma)} = 3/4$, and

$$b_2(1/2) = \frac{\sqrt{(1/2)(1/4)}}{1 - \sqrt{3/8}} \approx 0.912.$$

The tangent weight is

$$w_k = \sqrt{2} \left(\sqrt{\frac{2}{3}} \right)^k,$$

which decreases with photon number. The stochastic result says that no admissible output can have a larger expectation of this low-photon-number witness than the quantum-limited distribution, after accounting for its trace. The tangent inequality then converts precisely that one linear comparison into the nonlinear fidelity bound.

9 Imported facts versus the D.5 argument

Fact	Status
Vacuum-idler amplifier is least noisy in photon-number stochastic order	Imported from Guță–Matsumoto and used through Lemma D.3
One-mode conic inequality and its tensorization	Proved in Lemmas D.3 and D.4
Scalar tangent inequality (11)	Proved directly; its nonnegative gap is (12)
Factorization and monotonicity of the tangent weight	Checked directly from the thermal probabilities
Sharp fidelity bound and equality case	Proved in Lemma D.5

10 Dependencies and role in the paper

The full logical chain is

$$\text{Lemma D.1} + \text{Guță–Matsumoto} \longrightarrow \text{Lemma D.3} \longrightarrow \text{Lemma D.4} \longrightarrow \text{Lemma D.5}.$$

For Theorem 3.4, the Følner reduction first replaces the flat-prior problem by an exactly displacement-covariant map evaluated at the origin. Phase twirling supplies the phase covariance required here and cannot decrease the origin fidelity. Lemma D.5 then gives the upper bound

$$\prod_{i < j} b_\gamma(q_{ij}) = B_{\text{orb}}(\gamma, p).$$

The product quantum-limited amplifier gives the matching lower bound.

The hybrid converse later uses Lemma D.4 directly rather than D.5: its branchwise stochastic witness is combined with a separate classical Gaussian witness. Thus D.5 is the final step for the purely orbital theorem, while D.4 is the reusable stochastic component.

The minimal prerequisites are root fidelity for commuting states, multiplicativity, thermal distributions, phase covariance, and Lemma D.4. The only calculus idea is the tangent bound for a concave square root; identity (12) verifies it algebraically. The specialized least-noise theorem enters earlier through Lemma D.3, not in the tangent argument itself.